Technical Exercise - Position Book Application

Problem Statement

- As an investment bank we have the requirement to maintain the total quantity of a traded security**[1]** held at any point in time (this is referred as real time position).
- All the positions are stored in a Position Book, aggregated using the trading account and the security identifier. The Position Book also stores the details of all the events involving the trading account and security.
- The position book is updated whenever a Trade Event comes into the system; a trade event can represent an operation to buy a security, sell a security or cancel a previously issued event.
- A trade event always carries an ID and the number of securities to trade (buy or sell). A cancel event is meant to cancel any trade event with the same event ID; as a consequence, the quantity and security Id of a cancellation event are meaningless.

The goal is to develop a Position Book system that can process trade events and from which a user can retrieve a particular position identified by a trading account and security. The system should keep the data in memory (not in a database or in the file system)

[1] A security is a tradable asset of any kind. Examples of securities are debt securities (such as banknotes, bonds and debentures), equity securities, (e. g., common stocks) and derivative contracts, such as forwards, futures, options and swaps.

Requirements

Create REST APIs in java(Feel free to use java based frameworks like Spring boot) which exposes the endpoints to process trade events and to retrieve a particular position identified by a trading account and security.

Consideration should be given to input validation.

Please ensure full and high quality test coverage.

If any assumptions are made, please include a readme.txt file in your project explaining your rationale.

Different sample input event streams are provided, with the relative outputs. Please use those to guide the logic that you develop.

Your completed solution should be uploaded to a publically available repository on GitHub. Once the interview process is complete, the repository should be deleted.

Suggestions

Please summarise the problem on paper before starting the exercise, and ask any relevant question to the interviewer.

Examples

1. Buying Securities

   Input

Trading Events
1 BUY ACC1 SEC1 100 2 BUY ACC1 SEC1 50

   Output

Positions	Positions Drilldown Details
	[id: 1, BUY, ACC1, SEC1, 100]
ACC1 SEC1 150	[id: 2, BUY, ACC1, SEC1, 50]

2. Buying different securities

   Input

Trading Events
3 BUY ACC1 SEC1 12
4 BUY ACC1 SECXYZ 50
5 BUY ACC2 SECXYZ 33
6 BUY ACC1 SEC1 20

Output

Positions	Positions Drilldown Details
ACC1 SEC1 32	[id: 3, BUY, ACC1, SEC1, 12]
	[id: 6, BUY, ACC1, SEC1, 20]
ACC1 SECXYZ 50	[id: 4, BUY, ACC1, SECXYZ, 50]
ACC2 SECXYZ 33	[id: 5, BUY, ACC2, SECXYZ, 33]

3. Buying and Selling securities

Input

Trading Events
10 BUY ACC1 SEC1 100
11 SELL ACC1 SEC1 50

Output

Positions	Positions Drilldown Details
ACC1 SEC1 50	[id: 10, BUY, ACC1, SEC1, 100]
	[id: 11, SELL, ACC1, SEC1, 50]

4. Cancelling events

Input

Trading Events
21 BUY ACC1 SEC1 100
21 CANCEL ACC1 SEC1 0
22 BUY ACC1 SEC1 5

Output

Positions	Positions Drilldown Details
ACC1 SEC1 5	[id: 21, BUY, ACC1, SEC1, 100]
	[id: 21, CANCEL, ACC1, SEC1, 0]
	[id: 22, BUY, ACC1, SEC1, 5]